

Algorithm 5: Task Management System

Exercise 5: Task Management System

Scenario:

You are developing a task management system where tasks need to be added, deleted, and traversed efficiently.

Steps:

Understand Linked Lists:

Explain the different types of linked lists (Singly Linked List, Doubly Linked List).

Setup:

Create a class Task with attributes like taskId, taskName, and status.

Implementation:

Implement a singly linked list to manage tasks.

Implement methods to add, search, traverse, and delete tasks in the linked list.

Analysis:

Analyze the time complexity of each operation.

Discuss the advantages of linked lists over arrays for dynamic data.